

Report
Of
**Cybersecurity Internship: Cyber Security in Government
Operations**

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfillment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: CS-23411520

Name of Student: ANKUR KUMAR RAJAK

Batch: 2026-27 (4CSE10)

2026-27

CANDIDATE'S DECLARATION

I, Ankur Kumar Rajak do hereby declare that the internship report titled “**Cybersecurity Internship: Cyber Security in Government Operations**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Ankur Kumar Rajak

Roll No. CS-23411530

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name: Ankur Kumar Rajak

Roll No.. CS-23411530

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	
2.	Acknowledgment	
3.	Internship Completion Certificate	
4.	Project Description 4 .1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof must be attached	
5.	Bibliography/References	

CHAPTER 3: INTERNSHIP CERTIFICATE, OFFER LETTER



Offer to Join APCSIP-2026 Internship

Dear Ankur R. Rajak

Intern ID: APCSIP/2026/151

Based on good performance in the online screening test conducted by us, and your bright academic credentials, you are invited to join the **Amroha Police Cyber Security Internship Program, 2026 (APCSIP-2026)** to be held from **11 June to 22 June 2026**. Congratulations on passing a rigorous selection procedure and making it to APCSIP-2026.

As a participant in this prestigious internship program, you are expected to maintain high standards of discipline and ethics throughout the internship and beyond. We expect that you will use the skills gained from this internship for growth of your career, betterment of law enforcement, and development of our nation.

We proudly welcome you aboard.

Let's work towards the vision of **Viksit Bharat 2047**. Together.

Jai Hind

(Lakhan Singh Yadav, IPS)
Superintendent of Police Amroha

4. PROJECT DESCRIPTION

4.1 Introduction

This report documents the work undertaken during a 10-day cybersecurity internship completed in June 2026 under the Amroha Police Cyber Security Internship Program (APCSIP-2026), conducted by Amroha Police (U.P.) under the guidance of Shri Lakhan Singh Yadav, Superintendent of Police.

This training was designed specifically to acquaint the participants with the practical aspects of working as an investigation officer of Cybersecurity in governance – something that is not covered just by academics or theoretically – which include vulnerability assessment and penetration testing (VAPT), bug bounty, mobile and digital forensics, OSINT, dark web intelligence gathering, blockchain and cryptocurrency security, Google Dorking and the legal framework that guides cybercrime investigations in India.

4.2 Organization Profile

Amroha Police operates a dedicated Cyber Security Cell responsible for investigating cybercrime within its jurisdiction, ranging from financial fraud and phishing to cryptocurrency-related offences and online exploitation. The **APCSIP-2026** Internship Program was organized by the above-mentioned cell in order to develop manpower of technically trained individuals as well as students who can assist in cybercrime investigation and cyber awareness campaigns. The program was conducted under the leadership of Shri Lakhan Singh Yadav (Superintendent of Police) and Ms. Anjali Kataria (Deputy Superintendent of Police)

4.3 Problem Statement

Cybercrime in India has grown rapidly in both volume and plning, spanning phishing and social-engineering attacks, mobile malware, dark-web-facilitated crime, and increasingly, cryptocurrency-based fraud such as pig-butcheringscams, honey trap, digital arrest and many more.

4.4 Project Objectives

Main aims of this internship were to:

- Get familiar with the complete process flow of Vulnerability Assessment and Penetration Testing (VAPT) which includes reconnaissance, exploitation, and reporting.
- Get hands-on experience of bug bounty hunting process flow which involves subdomain enumeration, analysis of JS files, types of web vulnerabilities such as noSQL injection, LFI, privilege escalation etc.
- Learn the basic concepts of mobile and digital forensics which involve Android app structure, static and dynamic analysis of APK file using tools like MobSF and finding any insecure permissions or exposed modules.
- Utilize OSINT methods to legally collect the publicly available information for investigation using search-engine dorking and advanced reconnaissance tools.
- Assist the district cybercrime division in creating public awareness about cyber security issues and help officers in the field level investigations wherever assistance was required.

4.5 Scope of the Project

The scope of the internship was training-and-exposure based rather than a single software deliverable. It covered six broad technical modules — VAPT, bug bounty/web security, mobile and digital forensics, OSINT, dark web intelligence, and blockchain/cryptocurrency security — delivered over 10 days through instructor-led sessions and guided hands-on labs. Every day new topic new person new experience person.

4.6 Technologies and Tools Used

- Web/App Security: Burp Suite (with MCP-server extension setup for AI-assisted testing workflows), MobSF for Android APK analysis.
- OSINT & Reconnaissance: search-engine dorking (Google dorks), Yandex and Shodan-style asset search engines.
- Dark Web & OPSEC: Tor Browser and onion routing concepts.
- Blockchain & Crypto Investigation: block explorers (e.g. Etherscan), wallet-clustering and fund-tracing platforms.
- CTF & BugBounty : Hacker one, how to use Claude for BugBounty and pen-testing.
- HuggingFace / open-source models (building/fine-tuning your own AI model locally)
Ollama (running your own local AI model for testing)

4.7 System Architecture

Rather than a single software system, the internship's technical content was organized around two recurring architectural models used throughout the sessions:

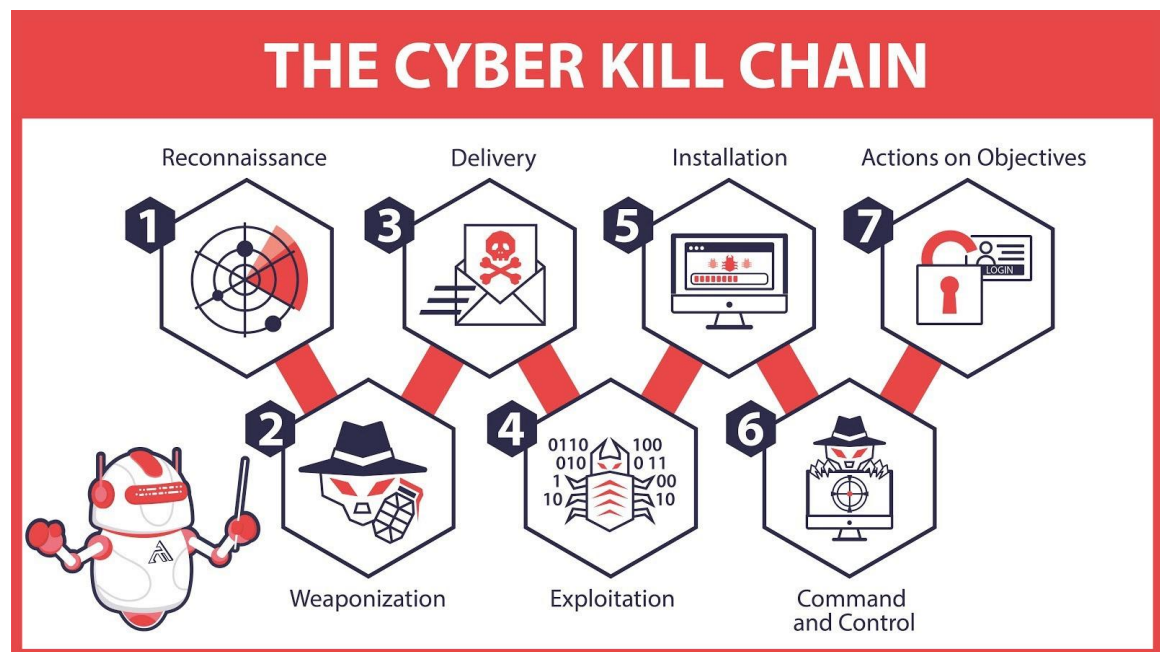


Figure 1: The VAPT/attack-lifecycle model

HOW DOES MONEY MULE FRAUD TYPICALLY WORK?

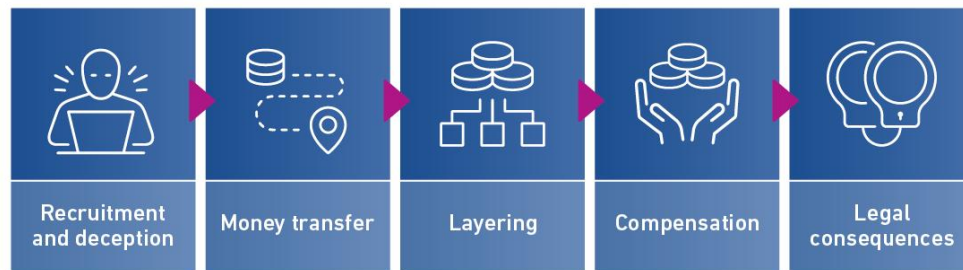


Figure 2 : How does a Mule account work

4.8 Methodology

The Internship was designed on a module-based approach where the session started with theory conducted by mentors and then followed by the practical lab session. For VAPT and Web security, the approach included setting up the Burp Suite with the MCP-server plugin for AI-based vulnerability finding followed by practical labs which included NoSql injection lab on a dummy application and local file inclusion in a production environment. For OSINT, it included practice in creating Google dork queries that reveal exposed admin panels and configuration files as well as performing asset searches through online platforms to list down all the assets related to the target company.

In the case of dark web and OPSEC modules, the progression from clean web to deep web/dark web, onion routing concepts, and the OPSEC techniques employed by the attackers such as using isolated VMs and VPN chaining were discussed from the perspective of an investigator for lawful surveillance purposes. The approach followed for the blockchain security module involved progression layer by layer in the technology stack with the help of case studies, which emphasized that attacks take place at the application/human level and not at the cryptographic layer.

Key learnings from the program included:

- Blockchain & Cryptocurrency Security
- Smart Contract Vulnerabilities
- Wallet Drainer & Phishing Attacks
- Social Engineering & Human-Layer Exploitation
- Pig Butchering and Investment Fraud Scams
- Deepfakes, Fake Job Offers & Digital Arrest Scams
- Blockchain Investigation Techniques
- Cyber Awareness — “Think · Verify · Then Click”

Session: Bug Bounty Hunting — by Yash Chavhan Sir

Session: Security Operations Centre (SOC) Readiness — by Sujeet Yadav Sir

Session: Mobile Forensics

Session: Android in the Pocket

Session: Research Paper Writing

Session: OSINT (Open-Source Intelligence) — by Ansh Sharma

Session: Malware exploitation methodologies — by Dr. Mr. OoPpSs

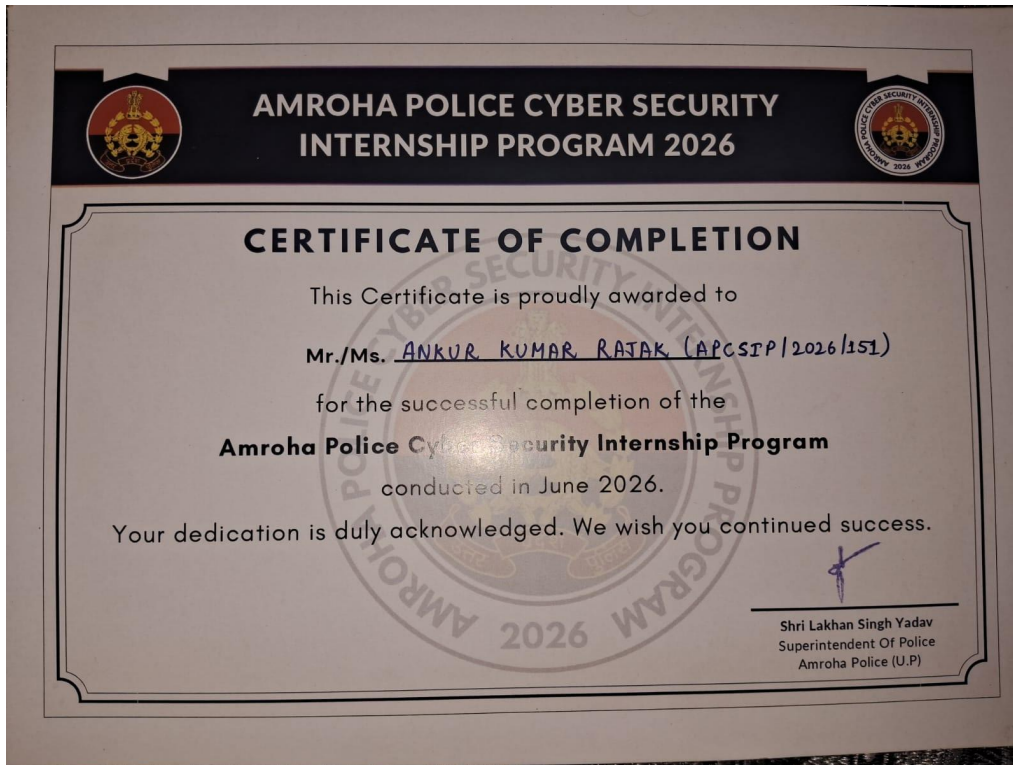
4.9 Expected Outcomes

- A working, practical understanding of the VAPT methodology and common web/mobile vulnerability classes.
- The ability to construct and use OSINT queries and asset-search platforms for lawful investigative reconnaissance.
- Familiarity with dark web structure and OPSEC concepts sufficient to support monitoring and awareness work.
- An understanding of how blockchain and cryptocurrency-related crimes occur and how on-chain data can be used to trace stolen funds to an off-ramp.
- Successful completion of the Amroha Police Cyber Security Internship Program 2026, evidenced by the completion certificate (APCSIP/2026/151).

4.10 Certificates of Completion and Other Communication Proof

The Internship Completion Certificate (APCSIP/2026/151), issued by Shri Lakhan Singh Yadav, Superintendent of Police, Amroha Police (U.P.)

INTERNSHIP CERTIFICATE, COMPLETION



5. BIBLIOGRAPHY / REFERENCES

1. MITRE ATT&CK Framework. Available at: <https://attack.mitre.org>
2. OWASP Foundation. OWASP Top Ten and Testing Guide. Available at: <https://owasp.org/>
3. Amroha Police Cyber Security Internship Program (APCSIP-2026) — training sessions and internal course material, June 2026.
4. Chainalysis. Crypto Crime and Tracing Methodology Reports. Available at: <https://www.chainalysis.com/>
5. MobSF (Mobile Security Framework) — official documentation. Available at: <https://mobsf.github.io/docs/>
6. PortSwigger Burp Suite Documentation. Available at: <https://portswigger.net/burp/documentation>
7. Etherscan — Ethereum Block Explorer. Available at: <https://etherscan.io/>
8. rekt.news — DeFi and crypto security incident archive. Available at: <https://rekt.news/>
9. experian - Money Mule : <https://www.experian.co.uk/blogs/latest-thinking/guide/money-mules/>
10. https://www.linkedin.com/posts/cop-anjalikataria_apcsipabr2026-internship-internships-ugcPost-7455892352960884738-Homl/

